

COURSE TITLE : BASIC INSTRUMENTATION ENGINEERING

COURSE CODE : 2211

COURSE CATEGORY : B

PERIODS/WEEK : 4

PERIODS/SEMESTER: 60

CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	Scope of Instrumentation engineering	15
2	Electrical circuits in instrumentation	15
3	Basic Electronics	15
4	Basic electronic circuits	15
TOTAL		60

COURSE OUTCOME

Module	G.O.	On completion of the study of this module the student will be able
1	1	To understand the basics of instrumentation engineering
	2	To understand measurements and instruments
	3	To understand static and dynamic characteristics
	4	To understand errors in measurement
2	1	To understand Electrical network theorems
	2	To understand basics of magnetics
3	1	To understand basics of Electronics
4	1	To understand DC Power supply
	2	To understand field effect transistors
	3	To understand MOSFET

On completion of the study the student will be able

MODULE I SCOPE OF INSTRUMENTATION ENGINEERING

1.1 .0 To understand the basics of instrumentation engineering

- 1.1.1 To Explain the Historical development of instrumentation engineering
- 1.1.2 To describe the Generalized Instrumentation system - Primary sensing, Variable conversion, Variable manipulation, Data transmission, Data presentation elements with examples.

1.2 .0 To understand measurements and instruments

- 1.2.1 To describe the Concept of Measurement
- 1.2.2 To classify instruments such as Absolute and secondary Instruments, Deflection and Null Deflection Instruments, Analog and Digital Instruments
- 1.2.3 To Explain Indicating, Recording and Controlling Functions of Instruments

1.3.0 To understand static and dynamic characteristics

- 1.3.1 To Define static Characteristics
- 1.3.2 To define dynamic characteristics
- 1.3.3 To explain Static Characteristics –Accuracy, precision, Sensitivity, linearity, resolution, hysteresis, Reproducibility, Drift, etc with problems to calculate these parameters

1.4.0 To understand errors in measurement

- 1.4.1 To define error
- 1.4.2 To explain Limiting Error
- 1.4.3 To describe Relative Limiting Error
- 1.4.4 To explain Gross Error, systematic Error, Random Error.
- 1.4.5 To illustrate Open Loop and Closed Loop systems With Examples

MODULE II ELECTRICAL CIRCUITS IN INSTRUMENTATION

2.1.0 To understand Electrical network theorems

- 2.1.1 To explain superposition theorem and its application in D.C circuits using a circuit diagram.
- 2.1.2 To state Thevenin's Theorem.
- 2.1.3 To find Thevenin's equivalent circuit for a given network.
- 2.1.4 To State Norton's Theorem
- 2.1.5 To explain the procedure for finding Norton's equivalent circuit.
- 2.1.6 To state Reciprocity theorem.
- 2.1.7 To explain the maximum power transfer theorem and its applications in electrical and electronic circuits.
- 2.1.8 To compute various unknown values in electrical circuits using the above theorems

2.2.0 To understand basics of magnetism

- 2.2.1 To explain the relation between magnetism & Electricity
- 2.2.2 To compare magnetic and electric circuit.
- 2.2.3 To define the terms Magnetizing force, Flux, Flux density, permeability, intensity of magnetization and relation between these.

- 2.2.4 To explain self inductance and mutual inductance
- 2.2.5 To define Faraday's Laws of Electromagnetic Induction

MODULE III BASIC ELECTRONICS

3.1.0 To understand basics of Electronics

- 3.1.1 To differentiate active and passive components
- 3.1.2 To define semiconductors
- 3.1.3 To define Intrinsic and Extrinsic semiconductors
- 3.1.4 To explain PN junction diode
- 3.1.5 To describe forward and reverse characteristics of P-N junction diode.
- 3.1.6 To explain the working of PNP & NPN Transistors
- 3.1.7 To illustrate CE, CB, & CC Configurations
- 3.1.8 To find α , β , γ gain factors

MODULE IV BASIC ELECTRONIC CIRCUITS

4.1 .0 To understand DC Power supply

- 4.1.1 To illustrate the working of half wave rectifier
- 4.1.2 To explain the working of full wave rectifiers
- 4.1.3 To compare half wave, full wave and bridge rectifiers
- 4.1.4 To explain the working of shunt Capacitor filter
- 4.1.5 To describe the working of zener diode voltage regulator
- 4.1.6 To describe the working of DC Power supply

4.2.0 To understand Field Effect Transistors

- 4.2.1 To explain the basic structure of FET
- 4.2.2 To illustrate the operation of FET
- 4.2.3 To describe the characteristics of FET

4.3.0 To understand MOSFET

- 4.3.1 To explain the basic structure of MOSFET
- 4.3.2 To illustrate the operation of MOSFET
- 4.3.3 To describe the characteristics of MOSFET

CONTENT DETAILS

MODULE I

Historical development of instrumentation engineering - Generalized Instrumentation system : Primary sensing, Variable conversion, Data transmission, Variable manipulation, Data presentation elements with examples - Classify instruments such as Absolute and secondary Instruments , Deflection and Null deflection Instruments – Analog and Digital Instruments - Indicating ,Recording and Controlling Functions of Instruments-Units of various electrical and non electrical quantities - Standards of measurements- International standards, Primary standards, Secondary standards, Working Standards- Static Characteristics : Accuracy, Precision, Sensitivity, Linearity, Resolution, Hysteresis, Reproducibility, Drift, etc with problems to calculate these parameters - Limiting Error - Relative Limiting Error - Gross Error, systematic Error, Random Error-Open Loop and Closed Loop systems With Examples.

MODULE II

Superposition theorem and its application in D.C circuits using a circuit diagram - Thevenin's Theorem- Thevenin equivalent circuit for a given network - Norton's Theorem - procedure for finding Norton's equivalent circuit - Reciprocity theorem - Maximum power transfer theorem and its applications in electrical and electronic circuits - Compute various unknown values in electrical circuits using the above theorems-Relation between magnetism & Electricity - Compare magnetic and electric circuit.

Definition of terms like Magnetizing force, Flux, Flux density, permeability, intensity of magnetization and relation between these. Self inductance and mutual inductance -Faraday's Laws of Electromagnetic Induction.

MODULE III

Active and passive components - Semiconductors - Intrinsic and Extrinsic semiconductors - PN junction diode - Forward and Reverse characteristics of P-N junction diode - Working of PNP & NPN Transistors - CE,CB,& CC Configurations - α , β , γ gain factors

MODULE IV

Working of Half wave rectifier - working of and Full wave rectifiers - Compare Half wave , Full wave and Bridge rectifiers. Working of shunt Capacitor filter - working of Zener diode voltage regulator - working of DC Power supply-Basic structure of FET – Operation of FET - Characteristics of FET - Basic structure of MOSFET - Operation of MOSFET- Characteristics of MOSFET of Instrumentation engineering

REFERENCES:

1. A.K.Sawhney, A course in Electrical and Electronic measurement and Instrumentation, Dhanpathray and Co
2. B.L Theraja, Electrical Technology Vol-I, S Chand and Co
3. V.K.Mehtha, Electrical and Electronic principles,
4. J.B Gupta, Electronic Devices and circuits, S.K.Kataria and Sons